Step 1 — The Code Index and the Context Builder — Hints

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Hint 1 — index the code, keep the file path

Doc08's RAG, over `.py` files:

- `build_code_store(repo_dir)`: `DirectoryLoader(repo_dir, glob="**/*.py", loader_cls=TextLoader)`, `RecursiveCharacterTextSplitter.split_documents(...)`, embed, `collection.add(...)`. The one thing that matters: keep each chunk's file path in `metadata["file"]` — that path is the citation.
- `retrieve_snippets(collection, question, k=4)`: `collection.query(...)` and build a list of `{"code": documents[i], "file": metadatas[i]["file"]}`.

Hint 2 — the two deterministic builders

Both take snippets and return something the rest of the pipeline uses — no model:

```
build_context(snippets):
    for each snippet: f"# File: {snippet['file']}\n{snippet['code']}"
    join the blocks with blank lines

cited_files(snippets):
    first-seen, de-duplicated list of snippet["file"]
```

The `# File:` header is what lets the model (Step 2) see which code came from which file, and `cited_files` is the "jump to source" list. Test both with hand-made snippets — two from `auth.py` → three `# File:` headers in the context but `auth.py` once in `cited_files`. The whole file set is in the Solution.